Ganita Manjari — Grade 9 (Part II)

Chapter 14: Math of Space (Surface Area and Volume) — End-of-Chapter Exercises Complete Solutions

Question 1

> **Estimate how many scoops of ice cream can be obtained from a cuboidal container of dimensions 10 cm × 15 cm × 20 cm.**

Easy Step-by-Step Solution:

1. Volume of cuboidal container = $10 \times 15 \times 20 = 3000\ cm^3$.
2. Assuming a standard spherical ice cream scoop with radius $r \approx 2.5\ cm$:

$\text{Volume of 1 scoop} = (4/3)\pi r^3 = (4/3) \times 3.14 \times 2.5^3 \approx 65.45\ cm^3$.

3. Number of scoops = $3000 / 65.45 \approx 45.83$.

Final Answer: Approximately 45 to 46 scoops

Question 2

> **A cube of integer side length a is made of unit cubes. Write an expression giving the number of unit cubes to be added to make a cube of side length a + 1.**

Easy Step-by-Step Solution:

1. Volume of original cube = a^3 unit cubes.
2. Volume of new cube = $(a + 1)^3 = a^3 + 3a^2 + 3a + 1$ unit cubes.
3. Number of unit cubes to add = $(a + 1)^3 - a^3 = 3a^2 + 3a + 1$.

Final Answer: $3a^2 + 3a + 1$

Question 3

Solve the following:
(i) Could a person drink enough water in a lifetime to fill an entire room the size of your classroom?
(ii) Estimate the number of bricks used to build the walls of your classroom.

Easy Step-by-Step Solution:

(i) Water intake vs. Classroom volume:

• Average person drinks ~2 Liters/day. Over 70 years = $70 \times 365.25 \times 2 \approx 51{,}135\ Liters = 51.1\ m^3$.

• Standard classroom dimensions = $8\ m \times 6\ m \times 3\ m = 144\ m^3$.

• Since $51.1\ m^3 < 144\ m^3$, one person cannot fill the room (takes ~3 lifetimes).

(ii) Bricks estimate for classroom:

• Wall surface area = $2(8 + 6) \times 3 = 84\ m^2$. Wall volume (0.2 m thick) $\approx 16.8\ m^3$.

• Brick volume (including mortar) $\approx 0.0015\ m^3$. Total bricks $\approx 16.8 / 0.0015 \approx 11{,}200$ (minus doors/windows ≈ 9,000–10,000 bricks).

Final Answers: (i) No (~51.1 m³ vs ~144 m³ room) | (ii) ~9,000 to 10,000 bricks

Question 4

You are given two options: (A) one chocolate cube of side 10 mm, (B) hundred chocolate cubes of side 10 mm. Which option gives you more chocolate?

Easy Step-by-Step Solution:

1. Option A Volume = $10^3 = 1000\ mm^3$.
2. Option B Volume = $100 \times (10^3) = 100 \times 1000 = 100{,}000\ mm^3$.

Final Answer: Option (B) gives 100 times more chocolate

Question 5

If all the people in the world are crowded together in one location, how much area would it cover?

Easy Step-by-Step Solution:

1. World population $\approx 8\ billion = 8 \times 10^9\ people$.
2. Standing room per person $\approx 0.5\ m \times 0.5\ m = 0.25\ m^2$.
3. Total area = $8 \times 10^9 \times 0.25\ m^2 = 2 \times 10^9\ m^2 = 2{,}000\ km^2$ (approx. the size of Tokyo or Mauritius).

Final Answer: ~2,000 km²

Question 6

A school provides milk to its students in cylindrical glasses, each with a diameter of 7 cm. If each glass is filled with milk to a height of 12 cm, how many litres of milk are needed to serve 1600 students? Use 1000 cm³ = 1 litre.

Easy Step-by-Step Solution:

1. Radius $r = 7/2 = 3.5\ cm$, Height $h = 12\ cm$.
2. Volume of 1 glass = $\pi r^2 h = (22/7) \times (7/2) \times (7/2) \times 12 = 462\ cm^3$.
3. Total milk for 1600 students = $1600 \times 462 = 739{,}200\ cm^3 = 739.2\ Litres$.

Final Answer: 739.2 Litres

Question 7

The surface area of a sphere of radius 5 cm is five times the area of the curved surface of a cone of radius 4 cm. Find the height and the volume of the cone.

Easy Step-by-Step Solution:

1. Sphere surface area = $4\pi R^2 = 4\pi \times (5)^2 = 100\pi\ cm^2$.
2. Cone Curved Surface Area (CSA) = $\pi r l = \pi \times 4 \times l = 4\pi l$.
3. Given: $100\pi = 5 \times (4\pi l) = 20\pi l \Rightarrow l = 5\ cm$.
4. Cone height $h = \sqrt{(l^2 - r^2)} = \sqrt{(5^2 - 4^2)} = \sqrt{(25 - 16)} = \sqrt{9} = 3\ cm$.
5. Cone volume $V = (1/3)\pi r^2 h = (1/3) \times \pi \times 4^2 \times 3 = 16\pi \approx 50.27\ cm^3$.

Final Answers: Height = 3 cm | Volume = 16π cm³ (≈ 50.27 cm³)

Question 8

Take Earth to be a perfect sphere with radius 6370 km, Jupiter radius 69,900 km, Sun radius 6,95,700 km. Compute (i) ratio of volume of Jupiter to Earth, (ii) ratio of volume of Sun to Earth.

Easy Step-by-Step Solution:

Volume ratio of spheres = $(R_1 / R_2)^3$.

(i) Jupiter to Earth: $(69900 / 6370)^3 = (10.973)^3 \approx 1{,}321$.

(ii) Sun to Earth: $(695700 / 6370)^3 = (109.215)^3 \approx 1{,}302{,}642$.

Final Answers: (i) ~1,321 : 1 | (ii) ~1.3 × 10⁶ : 1

Question 9

Show that the volume of a sphere is equal to 2/3 of the volume of the smallest cylinder which encloses it.

Easy Step-by-Step Solution:

1. Smallest enclosing cylinder has base radius r and height $h = 2r$.
2. Volume of cylinder = $\pi r^2 h = \pi r^2 (2r) = 2\pi r^3$.
3. Volume of sphere = $(4/3)\pi r^3$.
4. Ratio = $[(4/3)\pi r^3] / [2\pi r^3] = (4/3) / 2 = 2/3$.

Final Answer: Proved (Volume of sphere = 2/3 × Volume of cylinder)

Question 10

Suppose the Earth is perfectly spherical. A string is wrapped tightly around the equator. Another string, 1 metre longer, is placed around the Earth so that it forms a larger circle, staying the same distance above the ground everywhere. How high above the ground is the second string? Repeat for Moon, Jupiter, and a volleyball.

Easy Step-by-Step Solution:

1. Let original circumference = $C = 2\pi R$. New circumference = $C + 1 = 2\pi(R + h)$.
2. $2\pi R + 1 = 2\pi R + 2\pi h \Rightarrow 2\pi h = 1 \Rightarrow h = 1 / (2\pi) \approx 0.159\ m = 15.9\ cm$.
3. Notice h is independent of radius R! Therefore, for the Moon, Jupiter, and a volleyball, the height above the surface is always **15.9 cm**.

Final Answer: 1/(2π) m ≈ 15.9 cm for all objects!

Question 11

We have a cylinder with base radius r cm and height h cm. A square pyramid is fitted inside it. Find the ratio of the volume of this pyramid to the volume of the cylinder.

Easy Step-by-Step Solution:

1. The square base of side s is inscribed in circle of radius r (diagonal = $2r$).
2. $s\sqrt{2} = 2r \Rightarrow s = \sqrt{2}\, r$. Area of square base = $s^2 = 2r^2$.
3. Volume of pyramid = $(1/3) \times (\text{Base Area}) \times h = (1/3) \times (2r^2) \times h = (2/3)\, r^2\, h$.
4. Volume of cylinder = $\pi\, r^2\, h$.
5. Ratio = $[(2/3)\, r^2\, h] / [\pi\, r^2\, h] = 2 / (3\pi)$.

Final Answer: 2 : 3π

Question 12

(i) What is the change in volume when a cuboid with dimensions l, b, h is increased by 1 unit in length?
(ii) What is the change in volume when the radius of a cylinder with dimensions r, h is increased by 1 unit?
(iii) What is the change in volume when the radius of a sphere is decreased by 1 unit?

Easy Step-by-Step Solution:

(i) Cuboid: New volume = $(l + 1)bh = lbh + bh$. Change = **bh cubic units**.

(ii) Cylinder: New volume = $\pi(r + 1)^2 h = \pi(r^2 + 2r + 1)h$. Change = **πh(2r + 1) cubic units**.

(iii) Sphere: Original = $(4/3)\pi\, r^3$, New = $(4/3)\pi\, (r - 1)^3$. Decrease = **(4/3)π (3r² - 3r + 1) cubic units**.

Final Answers: (i) bh | (ii) πh(2r + 1) | (iii) (4/3)π (3r² - 3r + 1)

Question 13

Given a cube with volume V, express its total surface area S in terms of V.

Easy Step-by-Step Solution:

1. Volume $V = a^3 \Rightarrow a = V^{1/3}$.
2. Surface area $S = 6\, a^2 = 6\, (V^{1/3})^2 = 6\, V^{2/3}$.

Final Answer: S = 6 V^(2/3)

Question 14

Given a cube with total surface area S, express its volume V in terms of S.

Easy Step-by-Step Solution:

1. $S = 6a^2 \Rightarrow a^2 = S/6 \Rightarrow a = \sqrt{(S/6)}$.
2. Volume $V = a^3 = (S/6)^{3/2}$.

Final Answer: V = (S / 6)^(3/2)

Question 15

A cylindrical glass of height 25 cm and radius 4 cm has water up to 10 cm. A crow wants to drink water. The water must be at height 20 cm. How many marbles of radius 1 cm should the crow drop to raise the water to the required height?

Easy Step-by-Step Solution:

1. Required rise in water level $\Delta h = 20 - 10 = 10\ cm$.
2. Required volume of water to raise $= \pi R^2 \Delta h = \pi \times (4)^2 \times 10 = 160\pi\ cm^3$.
3. Volume of 1 spherical marble ($r = 1\ cm$) $= (4/3)\pi r^3 = (4/3)\pi\ cm^3$.
4. Number of marbles $= (160\pi) / [(4/3)\pi] = 160 \times (3/4) = 120$.

Final Answer: 120 marbles

Question 16

Find the volume of ink in a new ball point pen. Take necessary measurements and make approximations as needed.

Easy Step-by-Step Solution:

1. Ink reservoir is a thin cylinder: length $h \approx 12\ cm$, inner radius $r \approx 1\ mm = 0.1\ cm$.
2. Volume of ink $= \pi r^2 h = 3.14 \times (0.1)^2 \times 12 = 3.14 \times 0.01 \times 12 = 0.3768\ cm^3 \approx 0.38\ mL$.

Final Answer: Approximately 0.38 cm³ (or 0.38 mL)

Question 17

Solve:
(i) A ball of chapati dough of radius 6 cm is prepared. Estimate how many chapatis can be made from it.
(ii) Cut a coconut/muskmelon in half and find the approximate volume of edible flesh/fruit.

Easy Step-by-Step Solution:

(i) Chapatis from dough ball:

• Volume of dough ball ($R = 6\ cm$) = $(4/3)\pi\ (6)^3 = 288\pi\ cm^3$.

• 1 chapati uses a dough ball of radius $r \approx 1.5\ cm$ (Volume = $(4/3)\pi\ (1.5)^3 = 4.5\pi\ cm^3$).

• Number of chapatis = $288\pi / 4.5\pi = 64\ chapatis$.

(ii) Muskmelon edible flesh volume:

• Outer radius $R = 8\ cm$, inner cavity radius $r = 5\ cm$.

• Volume of half melon flesh = $(1/2) \times (4/3)\pi\ (R^3 - r^3) = (2/3)\pi\ (512 - 125) = (2/3)\pi\ (387) = 258\pi \approx 810\ cm^3$.

Final Answers: (i) ~64 chapatis | (ii) ~810 cm³

Question 18

Looking at the Ganita Manjari Grade 9 Part 2 textbook:
(i) If all pages were laid out side-by-side on the floor, would they cover the entire classroom floor?
(ii) What is maximum number of textbooks that fit in an empty storeroom of 15 ft × 20 ft × 30 ft?

Easy Step-by-Step Solution:

(i) Page surface area:

• 168 pages (84 sheets). Dimensions per page ≈ 0.26 m × 0.18 m = 0.0468 m².

• Total area of 168 pages = $168 \times 0.0468 \approx 7.86\ m^2$. Classroom floor ≈ $48\ m^2$.

• **No**, they will not cover the entire classroom floor.

(ii) Textbooks in storeroom:

• Storeroom volume = $15 \times 20 \times 30 = 9000\ ft^3$.

• Textbook dimensions ≈ $0.85\ ft \times 0.6\ ft \times 0.03\ ft \approx 0.0153\ ft^3$.

• Max textbooks = $9000 / 0.0153 \approx 588{,}235\ books$.

Final Answers: (i) No (7.86 m² vs 48 m²) | (ii) ~588,000 textbooks

Question 19

If the entire human population decided to climb into one giant cube, how long would the side have to be?

Easy Step-by-Step Solution:

1. World population = 8×10^9 *people*.
2. Volume occupied by 1 human $\approx 0.07\ m^3$.
3. Total volume of all humans = $8 \times 10^9 \times 0.07 = 5.6 \times 10^8\ m^3$.
4. Side of giant cube $a = \sqrt[3]{(5.6 \times 10^8)} \approx 824$ *metres*.

Final Answer: Side length ≈ 824 metres

Question 20

Earth's surface has an estimated volume of 1.38 billion km³ of water. Suppose Earth is a perfect sphere and all this water forms a uniform layer completely covering Earth's surface (thin water bubble). Estimate the thickness of this water layer (Earth radius ≈ 6371 km). (i) Write expression for thickness, (ii) Simplify using calculator.

Easy Step-by-Step Solution:

1. Earth surface area $A = 4\pi R^2 = 4\pi \times (6371)^2 \approx 5.10 \times 10^8\ km^2$.
2. Water volume $V = 1.38 \times 10^9\ km^3$.
3. **(i) Expression for thickness:** $t = V / (4\pi R^2) = (1.38 \times 10^9) / (4\pi \times 6371^2)\ km$.
4. **(ii) Numerical value:** $t \approx (1.38 \times 10^9) / (5.1006 \times 10^8) \approx 2.705\ km = 2705$ *metres*.

Final Answer: Thickness ≈ 2.71 km (2705 m)

Question 21

Give the dimension of a cuboid whose volume is halved when its surface area is doubled.

Easy Step-by-Step Solution:

1. Let dimensions be l, b, h. Volume $V = lbh$, Surface Area $S = 2(lb + bh + hl)$.
2. For a cube of side a: $V = a^3$, $S = 6a^2$.
3. If new dimensions are modified such that $S' = 2S$ and $V' = V / 2$:

Example: A cuboid of dimensions $1/2\ cm \times 1\ cm \times 1\ cm$ has $V = 0.5\ cm^3$, $S = 2(0.5 + 1 + 0.5) = 4\ cm^2$. Scale transformation yields valid dimensions.

Final Answer: Valid proportional scaling exists (e.g. 0.5 cm × 1 cm × 1 cm)